

PRANAV MURALI

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SUMMARY

Computer Science graduate student with a strong foundation in algorithms, object-oriented programming, full-stack development using Java and React. Experienced in building end-to-end applications, integrating APIs, and optimizing system performance. Thrives in collaborative, fast-paced environments and passionate about solving complex problems to create scalable, user-centric solutions

EDUCATION

ILLINOIS INSTITUTE OF TECHNOLOGY

Master of Science, Computer Science

Chicago, Illinois, USA

May 2025

Coursework: Advanced Database Organization, Mobile Development Application, Cloud Computing, Computer Networks
Introduction to Information Security, Big Data Technologies, Software Management, Parallel & Distributed Systems

JAIN UNIVERSITY

Bachelor of Technology, Information Science & Engineering

Bangalore, India

July 2022

Coursework: Software Engineering, Data Structures, Object-Oriented Programming and Design, Programming with Python, Automation Theory and Logic, Design & Analysis of Algorithms, Object-Oriented Design, Software Testing

SKILLS

Programming Languages – Python, Java, C, C++, R, Dart

Cloud & DevOps – AWS, GCP, Docker, Kubernetes, Kafka, CI/CD, Version Control, Cloud Computing

Databases & Data Science – MySQL, MongoDB, NoSQL, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, TensorFlow, OpenAI, ML Algorithms

Web & App Development – HTML, CSS, JavaScript, React, Flutter, RESTful APIs, Agile/Scrum

Tools & IDEs – IntelliJ IDEA, Eclipse, VS Code, Git, GitHub, Bash, Linux, MS Office, Tableau

EXPERIENCE

ILLINOIS INSTITUTE OF TECHNOLOGY

System Administrator

Chicago, Illinois, USA

March 2024 – May 2025

- Automated cloud resource allocation and user management using **Python** and **Bash**, streamlining the process and reducing manual tasks by 30%
- Provided technical support for students working in cloud environments, resolving configuration issues & ensuring seamless application deployment
- Maintained and monitored **cloud server infrastructure**, optimizing performance, uptime, and security for academic and development use

CONCENTRIX

Technical Engineer

Bangalore, India

February 2023 – August 2023

- Built scalable backend automation tools for **Adobe Creative Cloud** tools (Photoshop, Acrobat, InDesign, Premiere Pro) using **Java**, **JavaScript**, and **RESTful APIs**, streamlining creative processes and improving the efficiency of task execution between teams
- Developed a smart **chat & call routing system** for Adobe's OAC using **Java** and **AWS**, reducing customer wait times by **28%** and improving live agent connections
- Enhanced Adobe's customer analytics platform (Hendrix) using **Java**, **SQL**, and **Docker**, building reliable backend pipelines and integrating **CI/CD** workflows for real-time data access and operational efficiency

RIVERSAND TECHNOLOGIES

Full Stack Developer

Bangalore, India

July 2021 – October 2021

- Built scalable backend modules in **Java** and **Python** using **OOP principles** and design patterns; Designed microservices and integrated RESTful APIs to enhance system performance and cross-functional collaboration
- Utilized **Kafka** and **Apache Storm** to enhance real-time data processing, leveraging distributed systems for high availability and performance

PROJECTS

Full Stack Electronic Medical Records Platform

Chicago, Illinois, USA

- Built and deployed an EMR platform using **Java**, **Spring Boot**, **Hibernate** and **MySQL** to manage electronic medical records with role-based access, **RESTful APIs**, MVC architecture, Spring Data JPA, Maven, embedded **Tomcat** and **Postman** for modular design and database handling

ML-Enhanced Morse Code Authentication System for Accessible Security

Chicago, Illinois, USA

- Developed a contactless authentication system using **Python**, **OpenCV**, and **machine learning algorithms**, enabling users to input passwords via Morse code through eye blinks, with facial recognition and gaze tracking for enhanced accessibility and security.

DevSecOps Automation with GenAI-Driven Virtual Teams Using Scrum and Waterfall Methodology

Chicago, Illinois, USA

- Developed an end-to-end **DevSecOps** automation pipeline using **AutoGen**, **LangChain**, **CI/CD**, SAST, DAST, and IaC tools by simulating virtual team roles through **GenAI** across **Waterfall** and **Scrum** phases to enhance security, agility, and operational efficiency.

Infrastructure Performance Analysis using Shell & Virtualization Tools

Chicago, Illinois, USA

- Benchmarked performance across **Bare Metal** systems, **Docker**, **VMs** using **Shell scripting**, profiling tools to evaluate CPU efficiency, memory usage and startup latency under simulated workloads; provided insights into scalability, virtualization overhead for infrastructure optimization

HONORS

- Community** – NSS (National Social Service), Rotaract Bangalore South, Covid-19 Volunteer, Indian Student Association
- Interests** – Soccer Team captain, Winner of BMS Valorant Esports Open Tournament